Milestone 2 - VizKids

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1 Introduction

[ML 1] The dataset used in this project contains speeches delivered by member states during the United Nations General Assembly between 1970 and 2015. The dataset includes 7507 speeches from 199 countries, representing official statements by heads of state. The UN General Debate is the most visible annual forum in international diplomacy, yet the speeches themselves are rarely read outside of policy circles. The dataset is particularly valuable because these speeches provide a comprehensive overview of how governments publicly express their priorities and perspectives on international issues. Topics commonly discussed include security, international cooperation, development, climate change, and global conflicts. The target audience is of course students, but we want to talk to journalists and anyone curious about international relations who would never sit down and read 20 million words of UN transcripts.

2 Visualization Breakdown & Sketch

2.1 General Plots

These overview charts help you understand the dataset at a glance (size, shape, key themes) so that the country-level details on the map carry more meaning. These charts answer the basic questions you need before interpreting any single country's data.

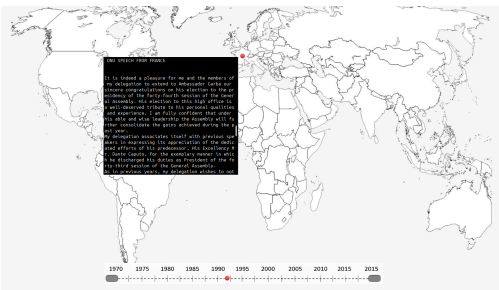
2.2 Information Map

This map will provide some basic information, such as displaying the full UN speech of a country for a given year. Specifically, we would have a timeline that the user can adjustand upon clicking (or hovering) a country the UN speech of that country in that year is displayed. Who speaks, how often and how long are critical elements.

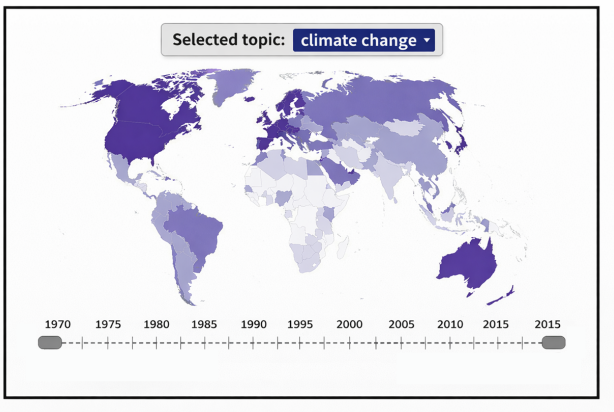
2.3 Search by Important Theme Map

This map will hold the most interesting insights. We would like to choose a topic and display a colored map reflecting the frequency at which the topic was mentioned throughout UN General Debates. The frequency of a topic will be determined by occurring words inherent to that topic. For example, if we choose "climate change", inherent words would be "climate", "emissions", "carbon", "sustainable", etc. The user can expand/shrink the timeline bar and see a modified version of the world. In this way a user can pick precise time intervals and visualize what the priorities of certain countries were at a specific time, how they evolved and where are they located. The words are very important: which topics dominate

the debate and how they rise and fall with geopolitical events (Cold War, 9/11, climate accords, refugee crises) ; but also the actors are very useful information: how individual countries differ from the global conversation, and which countries speak alike (blocs, alliances, regional patterns).



(a) Sketch for 2.1



(b) Sketch for 2.2

2.4 More advanced and potential extras

We could add in a way to compare debate of different countries during the same year. This could be a nice way to be able to get a geopolitical overview of the world at a given date for a selected list of countries. We could conduct cross analysis of speeches, for example retrieve speeches with a set of words, and compare them across different date or countries.

3 Risks and Mitigations

The performance is important: with 7 507 × 3 000 words it could be too much to process in-browser, so all the heavy processing is done offline and served as JSON. The other thing is the readiness: the information are stored per country, and small countries are often hidden from the big ones, even if small countries can have real impact and provide a lot of useful data. The magnifier lens (already implemented) addresses this, but could be improved.

4 Tools and Lectures

- **Frontend**: HTML, CSS, JavaScript (lectures 1-3) to build the core structure and logic of the application.
- **Data Processing**: Python(Pandas/Numpy) will be used for cleaning/filtering the UN speech dataset and extracting word frequencies, applying principles from lecture 4.1 (Data).
- **Interactive Visualization**: D3.js will be the primary tool for rendering the map and timeline. We will specifically apply techniques from lecture 4.2(D3.js) and lecture 5 (Interactions) to link the map and timeline dynamically. Lecture 8 (Maps) will be helpful for the creation of the maps.
- **Visual Perception**: we will use color scales based on lecture 6 (Perception of Colors) to ensure an appropriate use of colors based on the theme. Additionally, techniques introduced in lecture 5.1 such as *fisheye distortion* will be used to refine interaction.